A TAIL OF A QUASI-REGULAR SEQUENCE IS QUASI-REGULAR MODULO THE EARLIER ELEMENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Quasi-regularity via the associated graded module being polynomial. Used at the end of the counted proof (lines 17269-17273). Identifies the graded pieces with a polynomial algebra, which is what makes the intersection statement true.
- Deciding the general case by the case i = 1. Asserted at the first line of the counted proof (line 17254). Reduces the statement for arbitrary i to the case of one element, which is the case computed.

Lemma 0.1. *Let R be a ring. Let M be an R-module. Let $f_1, \ldots, f_c \in R$ be an M-quasi-regular sequence. For any i the sequence $\overline{f}_{i+1}, \ldots, \overline{f}_c$ of $\overline{R} = R/(f_1, \ldots, f_i)$ is an $\overline{M} = M/(f_1, \ldots, f_i)M$-quasi-regular sequence.*

Proof. It suffices to prove this for $i = 1$. Set $\overline{J} = (\overline{f}_2, \ldots, \overline{f}_c) \subset \overline{R}$. Then

$$\begin{aligned} \overline{J}^n\overline{M}/\overline{J}^{n+1}\overline{M} &= (J^nM + f_1M)/(J^{n+1}M + f_1M) \\ &= J^nM/(J^{n+1}M + J^nM \cap f_1M). \end{aligned}$$

Thus, in order to prove the lemma it suffices to show that $J^{n+1}M + J^nM \cap f_1M = J^{n+1}M + f_1J^{n-1}M$ because that will show that $\bigoplus_{n\geq 0} \overline{J}^n\overline{M}/\overline{J}^{n+1}\overline{M}$ is the quotient of $\bigoplus_{n\geq 0} J^nM/J^{n+1}M \cong M/JM[X_1, \ldots, X_c]$ by X_1. Actually, we have

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$J^nM \cap f_1M = f_1J^{n-1}M$. Namely, if $m \notin J^{n-1}M$, then $f_1m \notin J^nM$ because $\bigoplus J^nM/J^{n+1}M$ is the polynomial algebra $M/J[X_1, \ldots, X_c]$ by assumption. $\square$